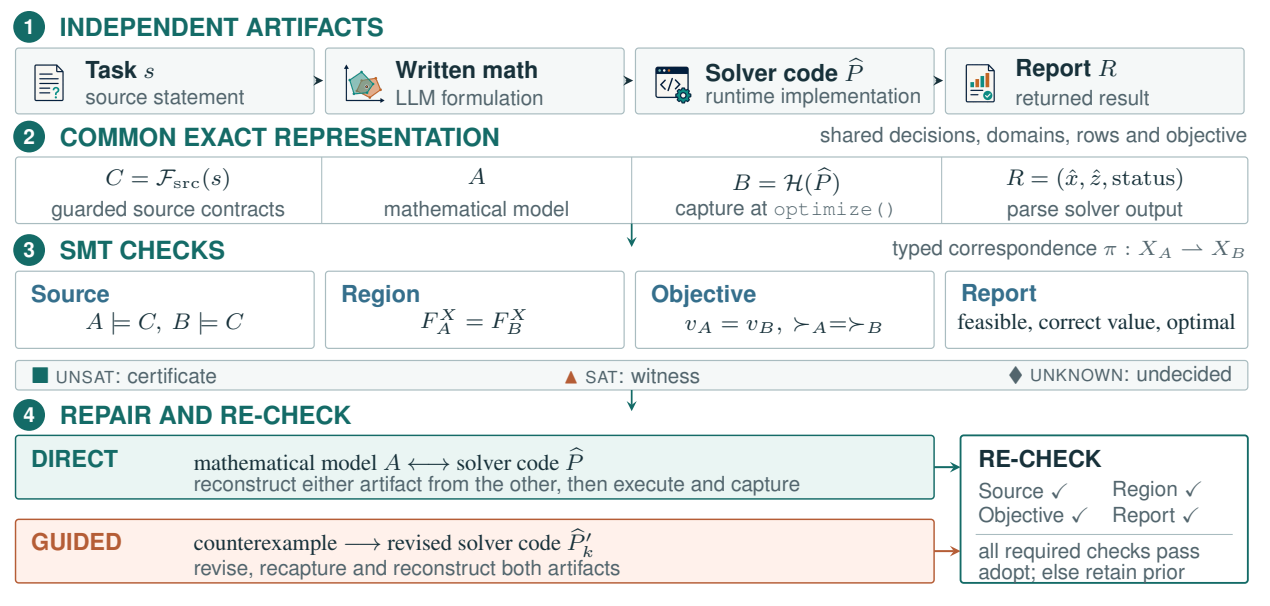

1 INDEPENDENT ARTIFACTS
Task s
source statement
Written math
LLM formulation
Solver code $\widehat{P}$
runtime implementation
Report R
returned result
2 COMMON EXACT REPRESENTATION
shared decisions, domains, rows and objective
$C = \mathcal{F}_{\mathrm{src}}(s)$
guarded source contracts
A
mathematical model
$B = \mathcal{H}(\widehat{P})$
capture at optimize()
$R = (\hat{x}, \hat{z}, \mathrm{status})$
parse solver output
3 SMT CHECKS
typed correspondence $\pi : X_A \rightharpoonup X_B$
Source
$A \models C,\ B \models C$
Region
$F_A^X = F_B^X$
Objective
$v_A = v_B,\ \succ_A = \succ_B$
Report
feasible, correct value, optimal
UNSAT: certificate
SAT: witness
UNKNOWN: undecided
4 REPAIR AND RE-CHECK
DIRECT
mathematical model $A \longleftrightarrow$ solver code $\widehat{P}$
reconstruct either artifact from the other, then execute and capture
GUIDED
counterexample $\longrightarrow$ revised solver code $\widehat{P}'_k$
revise, recapture and reconstruct both artifacts
RE-CHECK
Source ✓
Region ✓
Objective ✓
Report ✓
all required checks pass
adopt; else retain prior